Name : Lingesh M

Reg No : 411625104034

Department : B.E CSE

Subject : PYTHON

Batch : 02

Password Strength Checker using Python

Aim

The aim of this program is to design and implement a **Password Strength Checker** using Python.

This program evaluates the strength of a password based on:

- Length
- Presence of uppercase letters
- Presence of lowercase letters
- Presence of digits
- Presence of special characters

Procedure Step 1: Accepting User Input

The program asks the user to enter a password using the input() function.

Step 2: Checking Password Length

The length of the password is calculated using:

len(password)

Step 3: Initializing Condition Flags

Four Boolean variables are initialized:

- has_upper → Checks uppercase letters
- has_lower → Checks lowercase letters
- has_digit → Checks numbers
- has_special → Checks special characters

Initially, all are set to False.

Step 4: Defining Special Characters

A string containing allowed special characters is defined:

"!@#$%^&*()-_=+[]{}|;:'\",.<>?/"

Step 5: Checking Each Character

A for loop iterates through each character in the password and checks:

- isupper() → for uppercase letters
- islower() → for lowercase letters
- isdigit() → for digits
- Membership in special characters list

If a condition is satisfied, the corresponding flag becomes True.

Step 6: Determining Password Strength

The program checks conditions using if-elif-else statements:

- If length < 6 → Weak
- If length ≥ 6 and contains letters but no digits → Medium
- If length ≥ 8 and contains uppercase, lowercase, digit, and special character → Strong
- Otherwise → Moderate

Step 7: Displaying Result

The program prints the password strength accordingly.

Code

```
password = input("Enter your password: ")

length = len(password)

has_upper = False
```

```
has_lower = False

has_digit = False

has_special = False

special_characters = "!@#$%^&*()-_=+[]{}|;:'\",.<>?/"

for char in password:

    if char.isupper():

     has_upper = True

  elif char.islower():

       has_lower = True

   elif char.isdigit():

   has_digit = True

   elif char in special_characters:

       has_special = True if

length < 6:

   print("Password Strength: Weak  ")

elif length >= 6 and (has_upper or has_lower) and not has_digit:

   print("Password Strength: Medium")

elif length >= 8 and has_upper and has_lower and has_digit and has_special:

   print("Password Strength: Strong") else:

   print("Password Strength: Moderate")
```

Result Output:

```
Enter your password: 123@ligHtYagami
Password Strength: Strong
```

RESULT:

The Password Strength Checker was successfully implemented using Python.

The program correctly evaluates password strength based on length and character combination.

It effectively demonstrates:

- Conditional Statements
- Looping
- Boolean Flags
- String Handling